

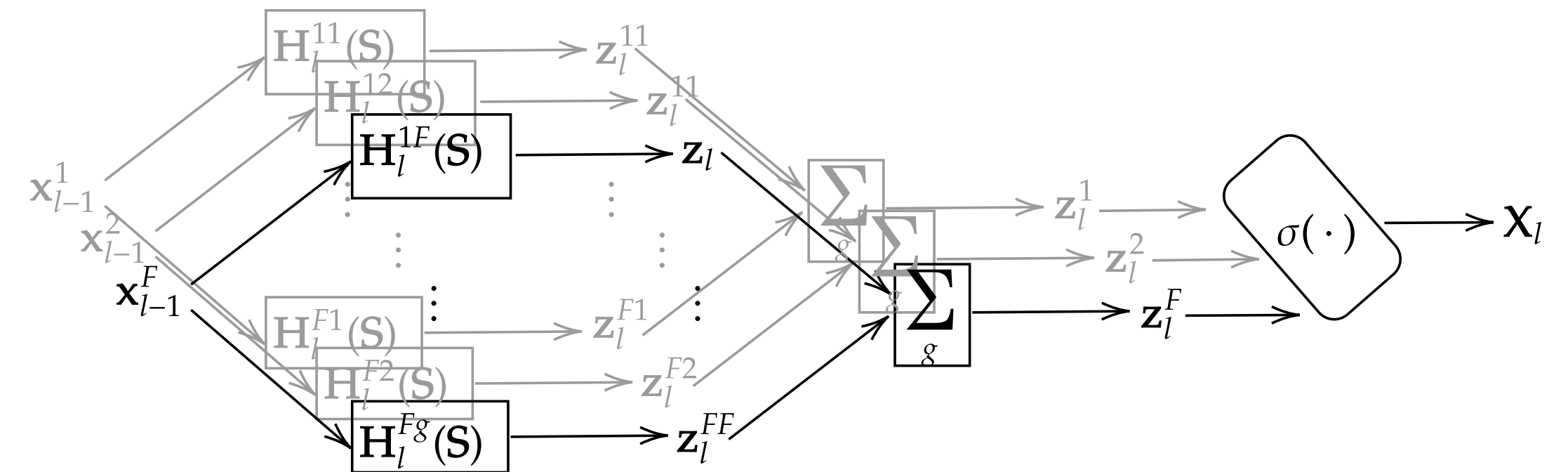
Simplicial CNNs

Edge-only interactions, Chebnet for edges

Linear

$$\mathbf{H} := \mathbf{H}(\mathbf{L}_d, \mathbf{L}_u; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \sum_{k=0}^{K_d} \alpha_k \mathbf{L}_d^k + \sum_{k=0}^{K_u} \beta_k \mathbf{L}_u^k$$

Non-Linear



Simplicial complex CNNs

Node-edge-triangle interactions

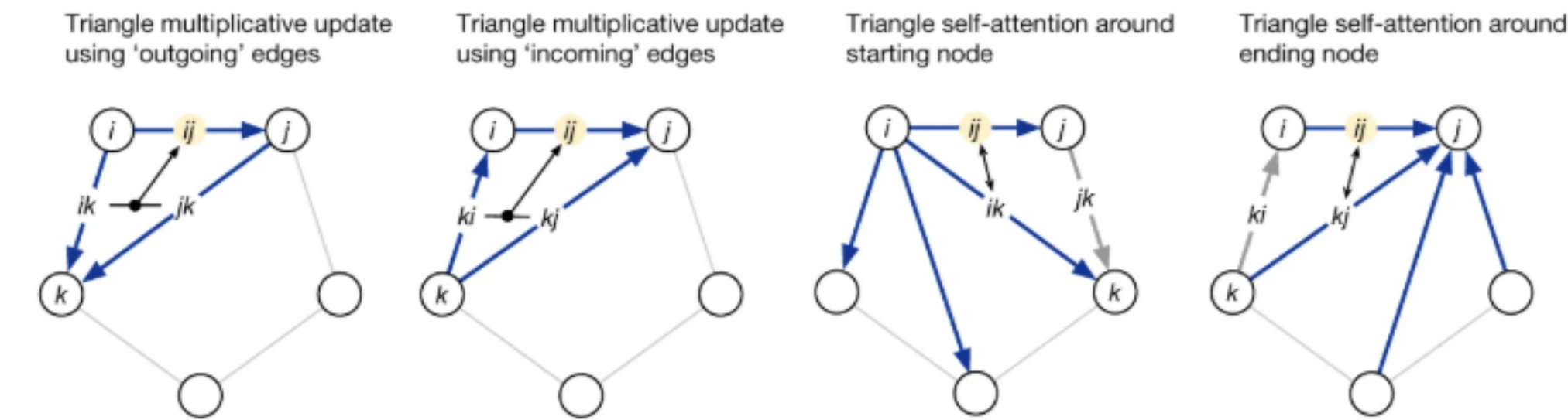
Node $\leftarrow$ node, edge
 Edge $\leftarrow$ node, edge, triangle
 Triangle $\leftarrow$ edge, triangle

Chebnet

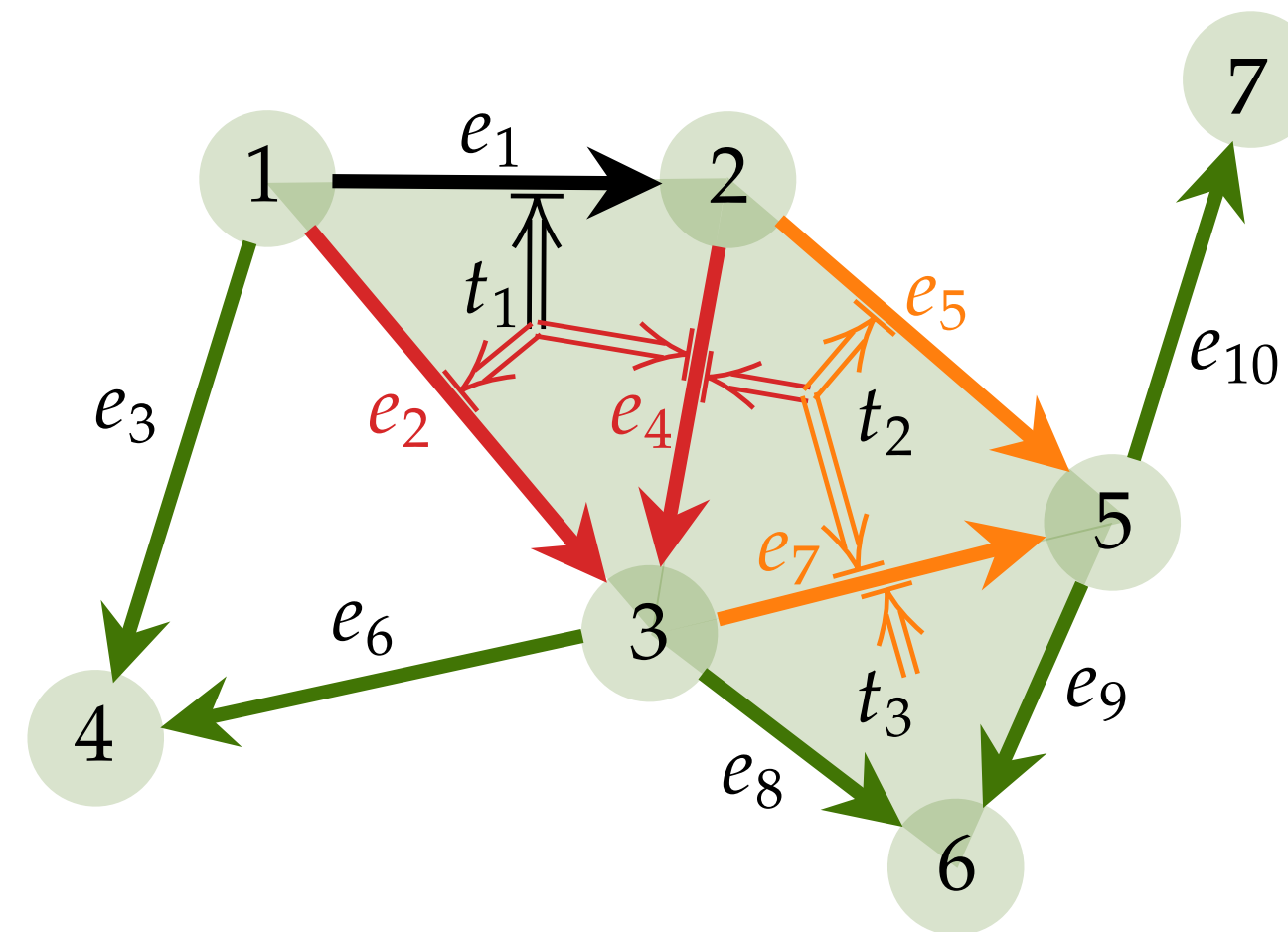
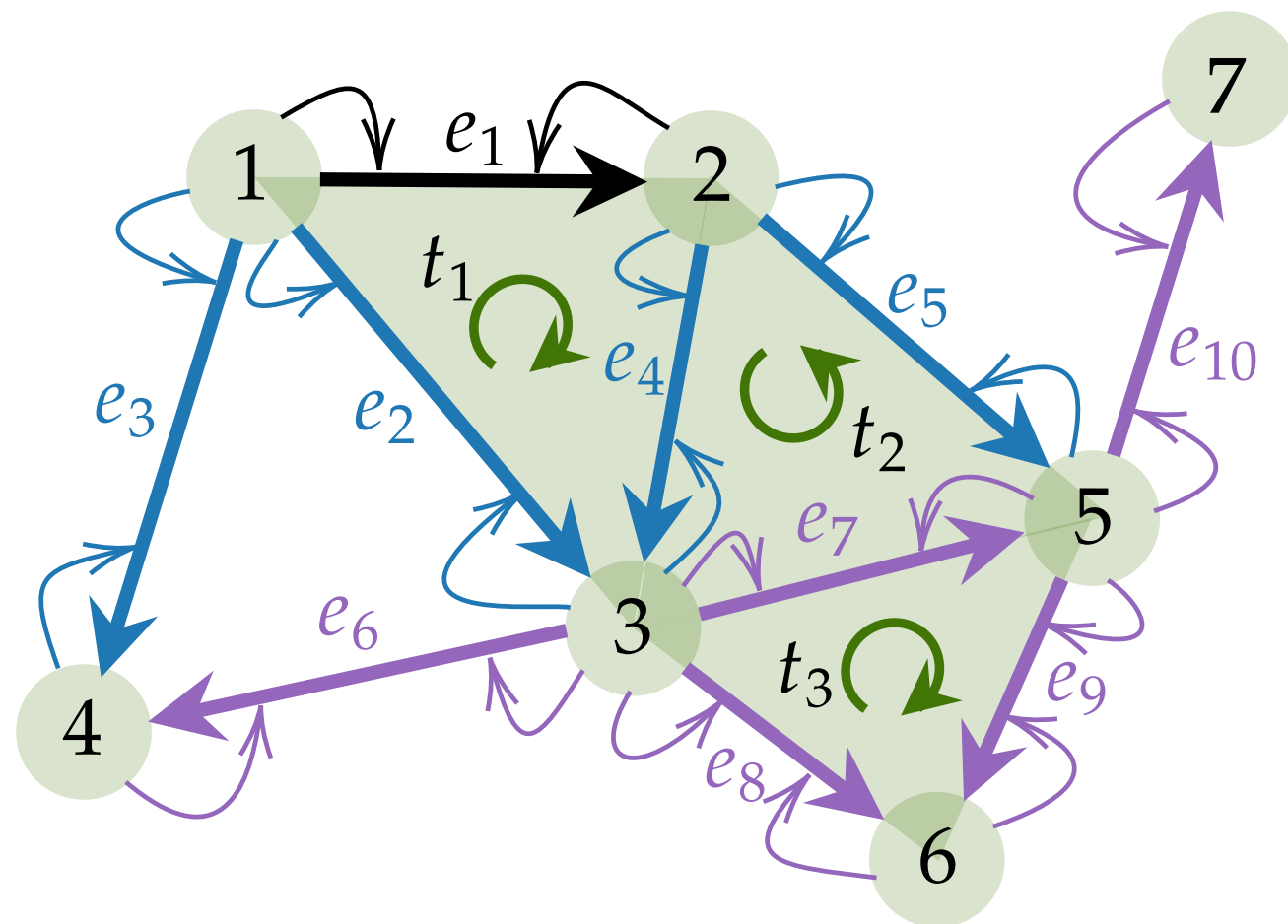
$$\mathbf{x}_0^l = \sigma(\mathbf{H}_0^l \mathbf{x}_0^{l-1} + \mathbf{H}_{0,u}^l \mathbf{B}_1 \mathbf{x}_1^{l-1})$$

$$\mathbf{x}_1^l = \sigma(\mathbf{H}_{1,d}^l \mathbf{B}_1^\top \mathbf{x}_0^{l-1} + \mathbf{H}_1^l \mathbf{x}_1^{l-1} + \mathbf{H}_{1,u}^l \mathbf{B}_2 \mathbf{x}_2^{l-1})$$

$$\mathbf{x}_2^l = \sigma(\mathbf{H}_{2,d}^l \mathbf{B}_2^\top \mathbf{x}_1^{l-1} + \mathbf{H}_2^l \mathbf{x}_2^{l-1})$$



AlphaFold2 paper Fig 3.
 Core element in AlphaFold3/4



Properties: locality, symmetry
 Dirichlet energy perspective
 Hodge-invariant
 Stability to weights perturbations
 Extensions:

- message-passing
- Attention
- E(3)-equivariance

Convolution based (Ebli et al. 2020; Roddenberry et al. 2021; Yang et al. 2022, 2023)
 Message passing (Bodnar et al. 2021)